

Solution 11
Iterative Optimization

Exercise 1

1. We have

$$f(x) = -e^{-x^2} \quad \frac{\partial f(x)}{\partial x} = 2e^{-x^2}x \quad (1)$$

The first iteration will be (See Lec. 11, slide 28)

$$\begin{aligned} x_1 &= x_0 - h \cdot (2e^{-x_0^2}x_0) \approx 0.0980 \\ x_2 &\approx -0.0961 \end{aligned} \quad (2)$$

2. First, we can calculate

$$\frac{\partial^2 f(x)}{\partial x^2} = e^{-x^2}(2 - 4x^2) \quad (3)$$

The step for each iteration will be

$$\Delta x_n = -\frac{\partial^2 f(x)}{\partial x^2} \frac{\partial f(x)}{\partial x} = -\frac{2e^{-x^2}x}{e^{-x^2}(2 - 4x^2)} = -\frac{2x}{(2 - 4x^2)} = -\frac{x}{(1 - 2x^2)} \quad (4)$$

We have

$$\begin{aligned} x_1 &= x_0 + \Delta x_0 = x_0 - \frac{x_0}{(1 - 2x_0^2)} \approx 0.0020 \\ x_2 &\approx -1.7 \cdot 10^{-8} \end{aligned} \quad (5)$$

3. See Matlab code.
4. If $\nabla f(x)^\top \Delta x < 0$, then there exists a step size $h > 0$ which is a descent direction (Lec. 11, slide 26). We can calculate that

$$\begin{aligned} \frac{\partial f(x)}{\partial x} \Delta x &< 0 \\ \Leftrightarrow \frac{-2e^{-x^2}x^2}{(1 - 2x^2)} &< 0 \end{aligned}$$

Since the nominator is always negative, the inequality can only be fulfilled if

$$1 - 2x^2 > 0 \Leftrightarrow |x| < \frac{1}{\sqrt{2}}.$$

Hence, $\frac{\partial f(x)}{\partial x} \Delta x < 0$ when $x \in \{x \in \mathbb{R} \mid -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}\}$.

5. The step for the gradient method is given by $\Delta x = -h \frac{\partial f(x)}{\partial x}$. Similar to the previous exercise, have that

$$\begin{aligned} \frac{\partial f(x)}{\partial x} \Delta x &< 0 \\ \Leftrightarrow -\left\| \frac{\partial f(x)}{\partial x} \right\|^2 &< 0 \end{aligned}$$

holds for all $x \in \mathbb{R}$. Thus, there always exists a step size h that leads to $f(x + \Delta x) < f(x)$.

6. See Matlab code.

Exercise 2

See Matlab code